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Research paper

Progressive Enhancement Dehazing for object detection in extreme weather

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ABSTRACT

Existing object detection technologies have made significant advances in perception systems for autonomous driving. However, accurately detecting objects in extreme weather conditions, especially fog, remains a significant challenge. On the one hand, current methods struggle to balance image enhancement and object detection, leading to the neglect of essential information that could improve detection accuracy. On the other hand, the lack of mature datasets in this area limits many studies to synthetic fog data generated from the prior-based atmospheric scattering model, which inevitably restricts further performance improvements. To address these challenges, we propose the **P**rogressive **E**nhancement **D**ehazing **Y**ou **O**nly **L**ook **O**nce (PED-YOLO) method, which processes images in real time under foggy conditions to improve object detection. We design a novel progressive image processing module that follows a unique paradigm of progressive supervised learning, which gradually processes the image from small size to large size, to effectively overcome the challenge of processing large-sized images at once. Moreover, we develop a small convolutional network module that focuses on each channel of the image and enables more accurate adaptive prediction of the parameters of the filters. In addition, we develop a novel style transfer model to generate simulated fog images that more closely resemble real fog images and train them together to bring the synthetic fog domain closer to the real fog domain and improve the generalizability. We evaluate our method extensively on several popular datasets, and the experimental results show the superior performance of PED-YOLO, highlighting its potential to advance autonomous driving.

1. Introduction

In recent years, significant breakthroughs have been made in autonomous driving technology (Ulusoy et al., 2023; Cheng et al., 2022; Perumal et al., 2021). This is mainly due to the continuous progress in object detection technology (Guo et al., 2023; Liu et al., 2024), which has driven the rapid development of perception systems for autonomous driving. Although existing object detection methods (Sun et al., 2021; Li et al., 2023) have shown excellent performance in clean images (Deng et al., 2009; Everingham et al., 2010), detection still needs improvement in extreme weather conditions, especially in fog, which is common in driving scenarios. According to the Road Traffic Accident Statistics Report by the Ministry of Public Security in China, approximately 10% of annual traffic accidents are directly attributed to extreme weather conditions such as rain, snow, and fog. Empirical studies (Das et al., 2022; Peng et al., 2021) corroborate that adverse weather conditions, notably fog, constitute a pivotal contributory factor to severe vehicular accidents. Foggy weather significantly impairs visibility and negatively affects the vehicle's vision system. This is mainly due to light scattering by water droplets and aerosol particles in the atmosphere, as shown in the first and third columns of Fig. 1, which reduce image contrast, distort colors, and lead to loss of object detail (He et al., 2010). All these factors together affect the performance of the object detection algorithm.

To mitigate the negative effects of foggy environments, there are two categories: (1) Two-stage models: Two-stage models use a defogging network (Dong et al., 2020; Li et al., 2017; Cai et al., 2016; Li et al., 2021) to pre-process the foggy image for detection. GridDeHaze (Liu

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Fig. 1. Intuitive cases explaining the problems of object detection in foggy conditions. We compare four versions: clean data (column 1), synthetic fog data created by the atmospheric scattering model (column 2), real fog data (column 3), and the defogged data after PED-YOLO (column 4).

et al., 2019a), for example, first denoises hazy images before detecting objects in them. Similarly, Sen et al. (2021) propose a U-shaped network to enhance input images, which are subsequently processed through an object detection pipeline. (2) Joint learning models: Joint learning models simultaneously train an image enhancement model and an object detection model to process images with poor visibility. More specifically, DSNet (Huang et al., 2020) uses two subnets with a shared feature extraction layer to simultaneously detect objects and improve image quality. DR-YOLO (Zhong et al., 2024) enhances foggy-condition object detection using a dehazing module and attention mechanism with prior knowledge integration.

However, machine vision is distinctly different from human vision, our appreciation for high-quality, defogged images does not necessarily lead to better machine recognition performance. The two-stage model, which uses an excellent defogging network to first recover the image and then feed it into the detector, does not guarantee an overall improvement in object detection performance. In addition, the combination of defogging and detection methods may slow down the detection process due to the additional defogging task. Although the joint learning model can effectively address these issues, it is challenging to adjust the weights between detection and defog during training. Moreover, some works (Chen et al., 2018; Hnewa and Radha, 2021) use domain adaptation to bridge the gap between detection and denoising. They mainly focus on fitting the two distribution features and neglect the potential information obtained during the recovery process based on fog images, thus failing to provide more information to the object detector.

Although numerous methods have been developed to address the challenge of object detection in foggy conditions, two major challenges persist. Firstly, traditional defogging methodologies necessitated the dehazing of a single, large-sized foggy image in one operation, coupled with the expectation of achieving exceptional dehazing results. This requirement poses a significant challenge to the model, as it has to effectively process large-sized images while ensuring high-quality defogging in just one iteration; Secondly, there is a notable lack of a mature, real-world dataset for object detector training. Most existing studies rely on synthetic fog datasets generated using atmospheric scattering models (Narasimhan and Nayar, 2002). The disparity between synthetic and real fog leads to poor generalization of models in actual foggy environments. In fact, most methods of synthesizing fog even contravene Koschmieder's laws (Zhang et al., 2021), which do not accurately model the aerosol composition that causes the atmospheric visibility changes. As illustrated in the second and third columns of Fig. 1, atmospheric scattering models struggle to capture the optical properties of the atmosphere in real foggy weather. Consequently, trained object detectors on synthetic fog data are faced great challenges in achieving excellent results in actual fog scenarios.

To address the above challenges, we propose an innovative real-time end-to-end object detector, dubbed **P**rogressive **E**nhancement **D**ehazing based **Y**ou **O**nly **L**ook **On**ce (PED-YOLO), which includes four modules, named progressive image processing module, channel adaptive predictor, contrastive learning module, and object detection module. **First**, we propose the progressive image processing (PIP) module to conduct the task of dehazing, which includes several custom filters, e.g., a defog filter, a gamma filter, a tone filter, a contrast filter, and a white balance filter. Additionally, we introduce a progressive filter strategy that begins by generating low-resolution dehazed images and gradually increases the resolution. By simplifying the task difficulty and progressively enhancing the resolution, this strategy enables the model to achieve high-quality detailed dehazing, ultimately facilitating the object detection performance under foggy conditions. This staged learning paradigm not only aligns with the natural process of human visual perception when interpreting images but also facilitates more efficient optimization during model training. **Second**, we design the channel adaptive predictor (CAP) to predict hyperparameters for filter components in PIP by estimating the tri-channel information derived from images. **Third**, we employ the contrastive learning module to overcome the poor generalization of detectors in actual foggy environments. Specifically, we present a novel CycleGAN-based (Zhu et al., 2017) method (AS-GAN) to generate simulated data. We then combine the simulated data with synthetic data to promote the detection performance in actual foggy environments. **Fourth**, we integrate the real-time object detector YOLOv7 with the above three modules through a joint learning approach. **Finally**, our proposal is thoroughly evaluated on several datasets, including the Pascal Visual Object Classes (VOC) (Everingham et al., 2010), the Real-World Task-Driven Testing Set (RTTS) (Li et al., 2018), and Cityscapes (Sakaridis et al., 2018), with PED-YOLO achieving state-of-the-art performance, indicating its potential to considerably accelerate advancements in the realm of intelligent transportation systems.

The main contributions of our work are as follows:

- We introduce PED-YOLO, a real-time object detection model designed to improve detection performance in foggy conditions. Experimental results indicate that PED-YOLO achieves state-of-the-art performance among real-time detectors under such challenging conditions.
- We propose a novel Progressive Image Processing Module (PIP), which incorporates custom filters and employs a progressive strategy designed to align with human visual perception processes and facilitate more efficient optimization during training. Following a coarse-to-fine principle, the module first eliminates coarse haze structures at low resolutions and iteratively filters fine haze details at progressively higher resolutions.
- We develop a channel adaptation predictor (CAP) for predicting the parameters of filters in PIP. It extends single-channel estimation to multiple channels and captures the unique information of each channel. Multi-channel images contain more detailed color information to help defogging. In addition, the method introduces a channel attention mechanism to accurately predict parameters and improve filter performance.
- We design a new style transfer model (AS-GAN) with atmospheric scattering constraints to generate more realistic simulated fog data. The generalization of the object detector is improved by combining synthetic data with simulated data using contrastive learning paradigm.

To validate the performance of our approach for object detection in foggy conditions, we undertook an extensive assessment of the proposed model using both synthetic and real-world foggy datasets. The extensive experimental results show that our proposed method outperforms the accuracy of current state-of-the-art real-time object detectors in foggy conditions and the cascade of defogging and detection models.

The remainder of this paper is organized as follows: Section 2 presents related work, including object detection in foggy conditions, image adaption for fog removal, and contrastive learning for domain adaptation. Section 3 describes the proposed method for object detection in foggy images. Section 4 presents the comprehensive experimental results of our method compared with other methods. Section 5 concludes the work and discusses the limitations and future work.

2. Related work

In this section, we will review the related work on object detection in foggy conditions, image adaption and contrastive learning.

2.1. Object detection in foggy conditions

Object detection in foggy conditions presents a significant challenge in the fields of computer vision and intelligent autonomous vehicles. Research on this subject has been limited (Nguyen et al., 2018; Hahner et al., 2021). A common approach (Dong et al., 2020; Li et al., 2017; Liu et al., 2019a) involves pre-processing images using traditional dehazing or image enhancement methods, which were initially designed to eliminate fog and enhance image quality. However, improved image quality does not necessarily translate to better detection performance. Some prior-based techniques perform image enhancement and detection simultaneously to minimize the influence of fog information. Sindagi et al. (2020) proposed an unsupervised, prior-based domain-adversarial object detection framework for detecting objects under hazy and rainy conditions.

Some methods utilize domain adaptation to address this problem. Hnewa and Radha (2021) identified a field shift between images taken under normal and foggy conditions. They created a multi-scale domain adaptive YOLO that supports domain adaptation at different layers during the feature extraction phase. Liu et al. (2022) developed an end-to-end joint image adaptation and object detection network called IA-YOLO. This network uses a CNN-based predictor to learn hyperparameters of the filters within a fully differentiable image processing (DIP) module. However, the DIP module is required to produce the dehazing effect for large-scale images at once, while the predictor performs parameter prediction based on a single channel, which might be challenging and inadequate for image processing tasks. Zhong et al. (2024) designed DR-YOLO, a knowledge-guided network using Restoration Subnet Module (RSM) for dehazing feature guidance, a Relation Reasoning Attention Module (RRAM) using co-occurrence graphs to compensate for fog-degraded features, and an Adaptive Feature Fusion Module (AFFM) for enhanced foggy detection with real-time performance. In this study, we employ a combination of IA-YOLO's image processing module and YOLOv7 (IA-YOLOv7) as the baseline model to enhance its performance in foggy environments. In contrast to IA-YOLO, our PED-YOLO employs a progressive image restoration approach that refines details layer by layer and extends the focus from a single channel to all three channels of the image simultaneously, enabling more comprehensive information capture from each channel. This structure aids in adaptively suppressing the influence of fog information more comprehensively and progressively extracting latent target features from coarse-grained to fine-grained.

2.2. Image adaption

To effectively enhance images, some traditional methods adaptively compute image transformation parameters based on the corresponding image features (Polesel et al., 2000). For example, an adaptive filter (Polesel et al., 2000) is applied that selectively enhances contrast in image regions with a lot of detail by manipulating the contribution of the sharpening path, while ensuring that there is minimal or no sharpening in regions that are already smooth. This approach strikes a balance between enhancing image contrast and preserving smoothness, enriching the details of the image.

In pursuit of adaptive image enhancement, a color image correction method (Wang et al., 2019a) is proposed to improve the match of low illumination images. The method is based on the illumination reflection model (Fu et al., 2016), multiscale theory (Modin et al., 2019), and Weber–Fechner law (Algom, 2021). It can effectively improve the brightness and contrast of the image, reduce the effects of uneven illumination, and make the image clean, bright and natural by constructing correction functions and image fusion strategies. In addition, an underwater image enhancement model (Gao et al., 2019) based on the morphology and function of fishnet skin is proposed to address the degradation of underwater images caused by blur and uneven color distortion. The model uses color-sensitive horizontal cell feedback and red channel compensation to correct uneven color distortion and improve the edges and contrast of the output image using key adaptive parameters. Moreover, a luminance-based fusion strategy (Moghimi and Mohanna, 2022) is used to improve the color enhancement and color correction of the images. This method does not require any knowledge of the underwater conditions or the structure of the scene, and the evaluation shows that the technique works excellently.

More recently, the image adaptive 3D look-up tables (3D-LUTs) (Zeng et al., 2022) are applied to achieve fast and robust image enhancement. While 3D LUTs are commonly used to adjust the color and hue of photographs, they typically require manual adjustment. They are the first attempt to learn 3D LUTs from annotated data using pairwise or unpaired learning. More importantly, the 3D LUTs can be adapted to different images, allowing flexible image enhancement. Additionally, Cross-to-Merge Training (C2MT) (Zhang et al., 2024) enhances robustness through cross-divided co-teaching, federated averaging, and class balancing, enabling noisy-label learning across diverse noise conditions.

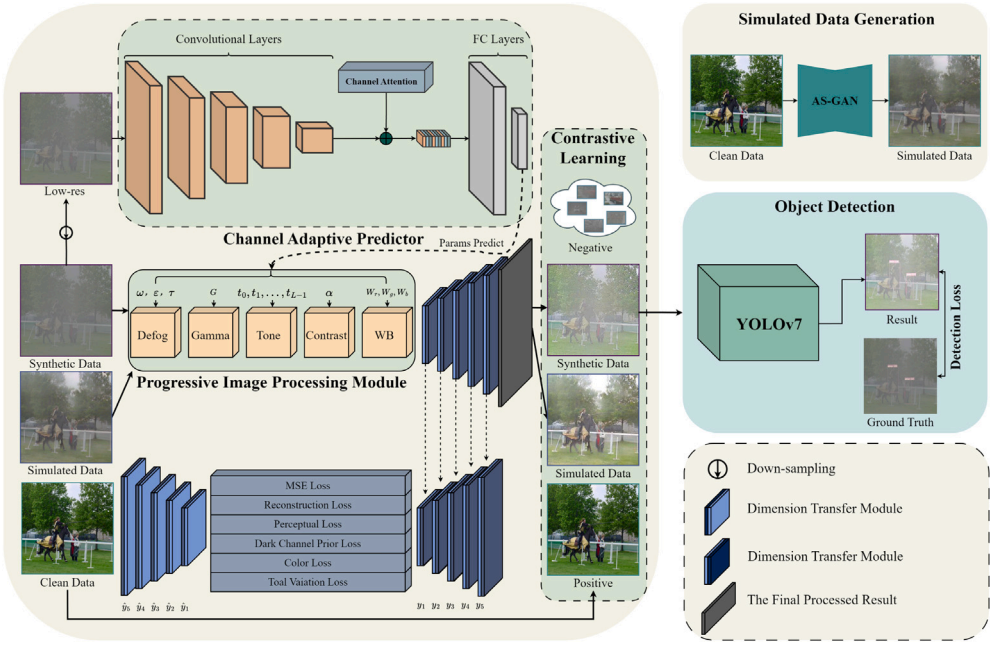

Fig. 2. Overall architecture of the proposed PED-YOLO for object detection under foggy conditions. PED-YOLO consists of simulated data generation, channel adaptive predictor (CAP), progressive image processing module, contrastive learning and object detection. Simulated data generation describes the simulated data is obtained by feeding the clean data into the style transfer model AS-GAN. The channel adaptive predictor utilizes the downsampled synthetic foggy image to predict the parameters of various filters in progressive image processing module. Five feature maps, y_1, y_2, y_3, y_4, and y_5, each of differing dimensions, are obtained by the input data through the progressive image processing module. These feature maps are then combined with five dimensions of the clean data, $\hat{y}_1$, $\hat{y}_2$, $\hat{y}_3$, $\hat{y}_4$, and $\hat{y}_5$, to calculate the multi-scale loss. Contrastive learning describes that the clean data and the processed simulated data serve as positive samples for the processed synthetic data. Five other synthetic data are randomly selected and utilized as negative samples to perform contrastive learning. The refined synthetic data is then input into YOLOv7 to detect objects.

2.3. Contrastive learning

Contrastive learning has been applied in many fields, such as image dehazing (Wu et al., 2021), context prediction (Santa Cruz et al., 2018), geometric prediction (Liu et al., 2019b), and image translation (Han et al., 2021). TUNIT (Baek et al., 2021) uses contrastive learning methods to manage images that share similar semantic structures. It understands how to separate these images from their associated domains and translate the input images into the intended domains. Regrettably, the semantic similarity assumption is not always applicable to every style transformation task, leading to a significant performance degradation in the acquired style representations. On the other hand, IEST (Chen et al., 2021b) overcomes significant hurdles in style transformation by cleverly solving problems such as color disharmony and repetitive patterns. IEST uses a special approach that merges the internal statistical data of each style image with the extensive information from large style databases to produce stylized images that are more rational and harmonious. Moreover, by introducing two types of contrast loss, the correlation between different stylizations is effectively managed, which increases the visual appeal of the images. In the context of IEST, contrastive learning serves as a tool for linking stylized images that share the same style, and draws its capabilities from the feature statistics of a pre-trained VGG (Simonyan and Zisserman, 2014). Meanwhile, PSSCL (Zhang et al., 2025) addresses image label noise through confidence-based subset refinement and semi-supervised learning (SSL), utilizing robust contrastive losses to enhance model robustness and achieving superior performance over existing benchmarks. Differently, we use contrastive learning to approximate the image domain of the synthetic fog image to the domain of the simulated fog image, further approximating the domain of the real fog image.

3. Proposed method

In this section, we will provide a detailed exposition of the proposed PED-YOLO framework. First, we delineate the architectural overview of the system. Next, we perform a rigorous introduce of its constituent modules.

3.1. Overview

Images taken during autonomous driving in foggy weather make object detection difficult due to light scattering from water droplets and aerosol particles in the atmosphere, reduced image contrast, color distortion, and loss of object detail. In addition, the lack of labeled data on real foggy days leads most studies to use synthetic foggy images based on physical models of atmospheric scattering. Since there is a significant domain gap between synthetic foggy images and real foggy images, models trained on synthetic foggy images cannot generalize well to real foggy images, which limits the performance of the model on real-world foggy images. To address these challenges, we propose an end-to-end image matching framework, which gradually extracts latent object features from coarse to fine levels and leverages domain contrastive learning strategies to break the bottleneck. The detailed architecture of our PED-YOLO is illustrated in Fig. 2.

To uncover more hidden information in foggy images, we propose the PIP module that applies a progressive image filtering method that refines details layer by layer and expands its focus from a single channel to all three channels of the image simultaneously. This approach allows for a more comprehensive capture of information in each channel. This structure helps to more effectively suppress the influence of fog information and extract latent target features from coarse-grained to fine-grained. To solve the generalization problem in real-fog environments, we design a novel style transfer model AS-GAN with physical constraint derived from the atmospheric scattering model, which can generate simulated fog images. By applying a contrastive learning paradigm, we combine synthetic fog images with simulated fog images to bring the domain of synthetic data closer to the domain of real fog. In this paper, we choose IA-YOLO (Liu et al., 2022) as the benchmark model. The filters embedded in this model are widely used, including a defog filter, a gamma filter, a tone filter, a contrast filter, and a

Table 1
The mapping function of pixel-wise filters.

Filter	Parameters	Mapping function
Gamma	G: gamma value	$P_o = P_i^G$
Tone	t_i: tone params	$P_o = (\mathcal{L}_{t_r}(r_i), \mathcal{L}_{t_g}(g_i), \mathcal{L}_{t_b}(b_i))$
Contrast	α: contrast value	$P_o = \alpha \cdot En(P_i) + (1-\alpha) \cdot P_i$
White Balance (WB)	W_r, W_g, W_b	$P_o = (W_r r_i, W_g r_g, W_b r_b)$

white balance filter (WB), which provide reliable technical support for our study. PED-YOLO consists of progressive image process module, channel adaptive predictor, contrastive learning on simulated data (simulated data generation and contrastive learning) and object detect. We will introduce each component in detail.

3.2. Progressive image processing module

Although numerous image processing methods have been developed (Mehra et al., 2020; Sahu et al., 2022), IA-YOLO (Liu et al., 2022) uses fully differentiable user-defined filters cascaded in a fixed sequence, to reduce computational overhead and meet the requirements of autonomous driving scenarios. A CNN-based predictor then learns the hyperparameters of each filter. This approach minimizes computational complexity, reduces resource requirements, and improves real-time performance. It also enables adaptive image processing to promote object detection performance in different scenes and environments. However, the filters in IA-YOLO focus only on the information in a single channel and filters large-sized images in one step, especially the major defog filter, which can leave out some critical information. The PIP module consists of the filters, progressive image process and multi-level loss. We divide the filters into two components, pixel-wise filters and defog filter. Next, we will introduce them in detail.

Pixel-wise Filters. In the pursuit of enhancing the visual appeal of images, we employ pixel-wise filters based on Liu et al. (2022). These filters execute specific operations that transform the input pixel values, denoted as $P_i = (r_i, g_i, b_i)$, into the output pixel values, represented as $P_o = (r_o, g_o, b_o)$. In this context, (r, g, b) correspond to the red, green, and blue color channels, respectively. The mapping function relationships utilized by these pixel-wise filters are detailed in Table 1. Moreover, the mapping functions are differentiable.

The architecture of the contrast filter integrates an input parameter that facilitates linear interpolation between the original image and its fully enhanced counterpart. As elucidated in Table 1, the formulation of $En(P_i)$ within the mapping function is defined as follows:

$$Lum(P_i) = 0.25r_i + 0.65g_i + 0.1b_i, \tag{1}$$

$$EnLum(P_i) = \frac{1}{2}(1 - \cos(\pi \times (Lum(P_i)))), \tag{2}$$

$$En(P_i) = P_i \times \frac{EnLum(P_i)}{Lum(P_i)}. \tag{3}$$

As in Liu et al. (2022) and Hu et al. (2018), we design the mapping function for the tone filter as a monotonic piecewise-linear function. We denote L as the number of parameters characterizing the tone filter, which are collectively represented by the set $t_0, t_1, \ldots, t_{L-1}$. The coordinates of points on the tone curve are defined as $(k/L, T_k/T_L)$, where $T_k = \sum_{l=0}^{k-1} t_l$. The tone filter's mapping function is formulated as follows:

$$P_o = \frac{1}{T_L} \sum_{j=0}^{L-1} clip(L \cdot P_i - j, 0, 1) t_k. \tag{4}$$

Defog filter. To achieve more efficient defogging, we adjust the core defog filter according to the atmospheric scattering model (Narasimhan and Nayar, 2002; McCartney, 1976). The definition of the atmospheric scattering model is presented below:

$$I(x) = J(x)t(x) + A[1 - t(x)], \tag{5}$$

where $J(x)$ represents the scene radiance (fog-free image), while $I(x)$ refers to the observed foggy image. A symbolizes the atmospheric light intensity, and $t(x) \in [0, 1]$ denotes the medium transmission map which is defined as the part of the light that is not scattered and arrives at the camera.

To obtain the clean image $J(x)$, it is essential to accurately determine the atmospheric light intensity A and the medium transmission map $t(x)$. This process begins with the calculation of the dark channel map, followed by the selection of the top 1000 brightest pixels. The value of A is then estimated by averaging the intensity of these selected pixels within the hazy image $I(x)$. Note that an image comprises three color channels: (r, g, b). Subsequently, as delineated in Eq. (5), we can derive that:

$$\frac{I^C(x)}{A^C} = t(x)\frac{J^C(x)}{A^C} + 1 - t(x), \tag{6}$$

where $C \in (r, g, b)$ represents the color channel. Then, we calculate the dark channel on both side of Eq. (6) by assuming the medium transmission map $t(x)$ is constant in a local patch $\Omega(x)$ as follows (He et al., 2010):

$$\min_{C, y \in \Omega(x)} \frac{I^C(y)}{A^C} = \min_{C, y \in \Omega(x)} [t(x)\frac{J^C(y)}{A^C} + 1 - t(x)], \tag{7}$$

where $\Omega(x)$ is a local patch centered at x, whose size is 15×15. Since the medium transmission map $t(x)$ remains constant within a local patch $\Omega(x)$, it can be extracted from the minimization operator.

$$\min_{C, y \in \Omega(x)} \frac{I^C(y)}{A^C} = t(x) \cdot \min_{C, y \in \Omega(x)} \frac{J^C(y)}{A^C} + 1 - t(x). \tag{8}$$

According to dark channel prior (He et al., 2010), in the local area of a haze-free image J(x), at least one color channel exhibits an intensity value that tends toward zero:

$$\min_{C, y \in \Omega(x)} \frac{J^C(y)}{A^C} = 0. \tag{9}$$

Therefore, the estimation equation is formulated as below:

$$t(x) = 1 - \min_{C, y \in \Omega(x)} \frac{I^C(y)}{A^C}. \tag{10}$$

The crux of the matter is to accurately obtain the atmospheric light intensity A and the medium transmission map $t(x)$. Nonetheless, existing estimation methods tend to primarily concentrate on the medium transmission map $t(x)$ while overlooking the impact of atmospheric light intensity A on the image. To address this issue, we simultaneously estimate the atmospheric light intensity A and the medium transmission map $t(x)$, as shown in Fig. 3. We additionally incorporate the parameters τ, ω and ε to adaptively control the degree of dehazing as follows:

$$t(x, \omega, \varepsilon, \tau) = 1 - \omega \min_{C, y \in \Omega(x)} \frac{\varepsilon I^C(y)}{\tau A^C}. \tag{11}$$

Since the operation of estimate $t(x)$ is differentiable, we can optimize the three parameters τ, ω and ε through backpropagation, thereby enhancing the effectiveness of the defog filter.

Progressive Image Process. We implement a progressive generation strategy to overcome the challenges prevalent in traditional defogging methods. These conventional approaches typically involve a single defogging operation on a large-sized foggy image, with an expectation of achieving outstanding defogging results. Our method addresses this by progressively filtering the foggy image, thereby facilitating more effective defogging. Adhering to the "coarse-to-fine" principle, this stepwise approach initially concentrates on filtering the coarse-grained structural features of the foggy content in the image (low-resolution) and then progressively refiltering the fine-grained details of the image (high-resolution) during the image processing stage, rather than attempting to defog large-sized images in one step. As depicted in the PIP module of Fig. 2, the output of filters is divided into y_1, y_2, y_3, y_4, and y_5. The resulting feature maps have dimensions of 40×40, 80×80, 160×160, 320×320, and 640×640. The clean image is

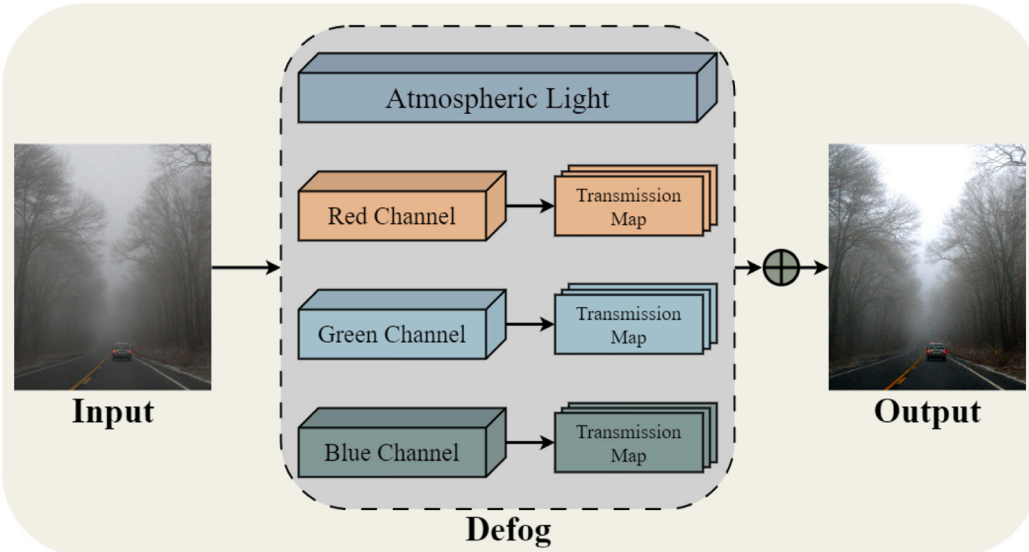

Fig. 3. Illustration of the defog filter. By estimating the atmospheric light and calculating the medium transmission map from the three channels of the input image, a more comprehensive dehazing function is achieved.

downsampling to match the corresponding feature map size, $\hat{y}_1$, $\hat{y}_2$, $\hat{y}_3$, $\hat{y}_4$, and $\hat{y}_5$, and then calculates the multi-scale loss.

Multi-scale Loss. Optimizing image visual fidelity is crucial to image dehazing. As such, we design six loss functions to achieve these objectives, including mean squared error (MSE) loss, reconstruction loss, perceptual loss (Johnson et al., 2016), dark channel prior (DCP) loss (He et al., 2010), color loss, and total variation loss, as shown in the bottom part of Fig. 2. More precisely, the mean squared error (MSE) loss ensures that the dehazed results are closer to the clean image across different scales. By summing up the MSE losses at various scales, we arrive at a formal definition as follows:

$$\mathcal{L}_{MSE} = \frac{1}{wh}\sum_{i=0}^{w-1}\sum_{j=0}^{h-1}(y(i,j)-\hat{y}(i,j))^2, \tag{12}$$

where w and h represent the width and height of the image, respectively. y denotes the synthetic data after PIP, and $\hat{y}$ denotes the clean data. Furthermore, during the dehazing process, we design an additional objective function to improve the model's robustness by mitigating the impact of outliers on image processing outcomes:

$$\mathcal{L}_{rec} = \left|y-\hat{y}\right|. \tag{13}$$

We use perceptual loss to more effectively learn high-level structural and textural information in images, ultimately producing images with improved visual quality and more natural textures as Johnson et al. (2016). The goal of perceptual loss is to minimize the $\mathcal{L}_1$ distance between the resulting feature map and the corresponding clean data on the activation map. The formal definition of perceptual loss is as follows:

$$\mathcal{L}_{per} = \sum_j \frac{\left|\psi_j(y)-\psi_j(\hat{y})\right|}{N_j}, \tag{14}$$

where ψ_j is the activation map from the jth layers of a pre-trained network (e.g., VGG19 (Simonyan and Zisserman, 2014)), N_j is the number of elements in ψ_j.

Based on He et al. (2010), we strengthen the dark channel constraint in the defogging task to minimize the disparity between the filtered feature map and the corresponding clean data. The formal definition of the priority loss function for the dark channel is presented as follows:

$$\mathcal{L}_{dcp} = \frac{1}{n}\sum_{z=1}^{n}\max_{p\in\Phi(z)}\left(\min_{c\in(R,G,B)} y_p^c\right), \tag{15}$$

where n represent the resolution of the feature map, with $\Phi(z)$ denoting the window centered on pixel z. y_p^c signifies the value of pixel p on the color channel c.

To improve image clarity and mitigate haze effects, we incorporate color loss and total variation loss into our approach. The color loss function is specifically designed to preserve color fidelity in the rectified image and minimize color distortion. The formal definition of the color loss function can be expressed as follows:

$$\mathcal{L}_{color} = MSE(G(y)_{blur}, G(\hat{y})_{blur}), \tag{16}$$

where $G()_{blur}$ denote the fuzzy representations obtained by applying a Gaussian filter.

In addition, we use a loss of total variation to obtain a smoother dehazing image that reduces fog while retaining image detail and edge information. This loss promotes the creation of dehazing images that maintain a balance between smoothness and the preservation of important details. Total variation loss is formally defined as follows:

$$\mathcal{L}_{tv} = \sum_{i=0}^{w-1}\sum_{j=0}^{h-1}\sqrt{(y(i+1,j)-y(i,j))^2+(y(i,j+1)-y(i,j))^2}, \tag{17}$$

where $(y_i)_{k,j}$ represents the pixel value located at row k and column j.

The progressive image process module's loss function comprises of these six individual loss components.

$$\begin{aligned}\mathcal{L}_{pip} = \sum_{k=0}^{4}\lambda_k \times (&\mathcal{L}_{tv}^k + \mathcal{L}_{color}^k + \mathcal{L}_{dcp}^k \\ &+ \mathcal{L}_{per}^k + \mathcal{L}_{rec}^k + \mathcal{L}_{MSE}^k),\end{aligned} \tag{18}$$

where $\lambda = (0.0001, 0.001, 0.01, 0.1, 1)$ represents the weights of different scale.

3.3. Channel adaptive predictor

To enhance overall image filter performance and accurately predict the hyperparameters of filter components, we adopt an approach that diverges from traditional parameter predictors. In this method, we fully acknowledge that each channel of an image may possess unique information, thereby expanding single-channel estimation to encompass multi-channel estimation. Compared to single-channel images, multi-channel images offer a more comprehensive representation of color and detail, which aids in effectively tackling image dehazing tasks.

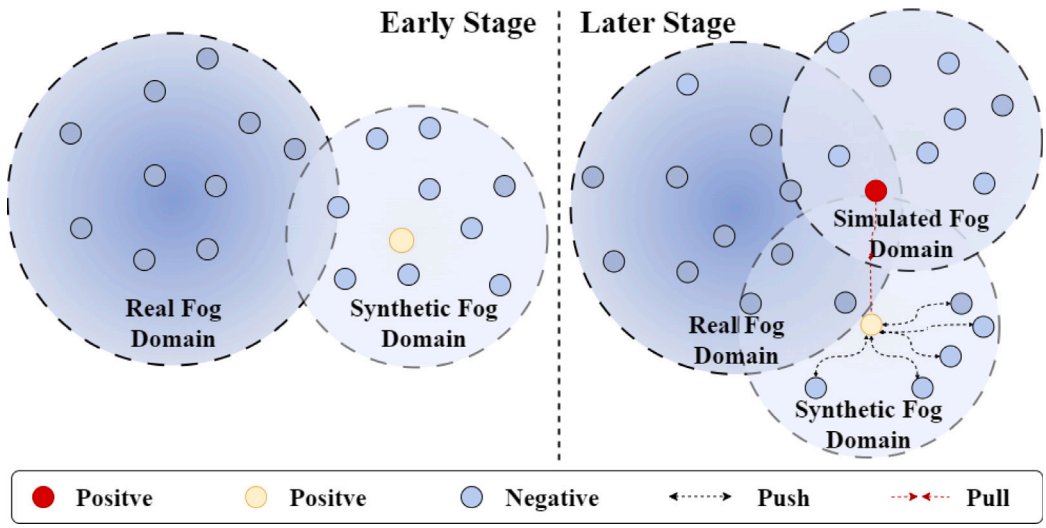

Fig. 4. Illustration of contrastive learning. We use simulated data to bridge the gap between the synthetic fog domain and the real fog domain.

In addition, we introduce a channel attention mechanism to focus on crucial information within different channels, allowing for more refined adaptive predicting the parameters of filters. This method efficiently leverages the channel information within images to promote the performance of the filters.

To more accurately predict the parameters of each filter, we take into account the overall content of the image, such as brightness, color, hue, and the extent of fog. In order to reduce computational demands, we downsample the image to the resolution of 256×256 using bilinear interpolation. The channel adaptive predictor consists of five convolution blocks, a channel attention layer, and a fully-connected layer. Each convolution block is composed of a 3×3 convolution layer and a Leaky ReLU activation function.

To adaptively accommodate input images of varying sizes, we employ an adaptive average pooling operation, which enhances the flexibility and compatibility of the predictor. Simultaneously, we integrate a channel attention mechanism to assign different weights to distinct channels, emphasizing important channel information and achieving finer-grained adaptive prediction parameter manipulation.

3.4. Contrastive learning on simulated data

Given the scarcity of labeled data in actual fog conditions, most studies have traditionally leveraged atmospheric scattering models (referenced as Eq. (5)) to create synthetic fog datasets. Although this method facilitates model adaptation to fog characteristics and improves object detection capabilities, reliance solely on synthetic data compromises the model's generalizability in actual fog scenarios. Addressing this limitation, our work implements a contrastive learning approach by integrating the new fog data (simulated data). First, we design a new style-transfer model (AS-GAN) to generate the simulated data, as shown for simulated data generation in Fig. 2. Then, we narrow the gap between the synthetic and real fog image domains by integrating simulated data, as depicted in Fig. 4. This strategy enhances the model's adaptability, overcoming previous limitations and significantly enhancing object detection accuracy in foggy environments. Next, we will introduce the simulated data generation and contrastive learning in detail.

Simulated Data Generation. In our study, we propose the AS-GAN, which is based on CycleGAN model, enhanced with well-defined atmospheric scattering constraints, to generate simulated images that closely mimic real fog conditions. The CycleGAN model is famous for the principle of cycle consistency, which builds a bridge between two different style domains. To this end, we introduce CycleGAN in the task of modeling the atmospheric properties of fog and impose a strong physical constraint by the atmospheric scattering model, as shown in Fig. 5. The overall Loss of AS-GAN can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{GAN}(N - F) + \mathcal{L}_{GAN}(F - N) + \mathcal{L}_{cyc} + \mathcal{L}_{con}, \tag{19}$$

where N and F represent the normal image and real fog image, respectively. $\mathcal{L}_{GAN}$ denotes the loss of GAN, $\mathcal{L}_{cyc}$ denotes the loss of the principle of cycle consistency, and $\mathcal{L}_{con}$ represents the loss of the physical constraint. Their specific definitions are as follows:

$$\mathcal{L}_{GAN}(N - F) = log D_F(F) + log(1 - D_F(Gen_{N-F}(N))), \tag{20}$$

$$\mathcal{L}_{GAN}(F - N) = log D_N(N) + log(1 - D_N(Gen_{F-N}(F))), \tag{21}$$

$$\begin{aligned}\mathcal{L}_{cyc} = &\left|Gen_{N-F}(Gen_{F-N}(F)) - F\right| + \\ &\left|Gen_{F-N}(Gen_{N-F}(N)) - N\right|,\end{aligned} \tag{22}$$

where Gen_{N-F} and Gen_{F-N} denote the generators from the normal image domain to the real fog image domain and the real fog image domain to the normal image domain, respectively. D_N and D_F denote the discriminators from normal image domain and real fog image domain, respectively.

The physical constraints are formulated as follows:

$$\mathcal{L}_{con} = MSE(Gen_{N-F}(N), P(N)), \tag{23}$$

where P denotes the atmospheric scattering model. Additionally, we utilize the MRFID (Liu et al., 2020) dataset, which comprises actual fog images with corresponding clean images, for model training. This approach enables AS-GAN to accurately learn the characteristics of real fog, thereby facilitating the creation of highly realistic simulated fog data. As shown in Fig. 6, we compared clean images, synthetic fog images, and simulated fog images to visualize the fog effect. The fog generated by AS-GAN is more natural than that generated by the atmospheric scattering model.

Contrastive Learning. We employ the contrastive learning to integrate two types of data, thereby enhancing the PED-YOLO's domain adaptability. In particular, we concurrently input synthetic data and simulated data into the PIP module, yielding outcomes y and y_s,

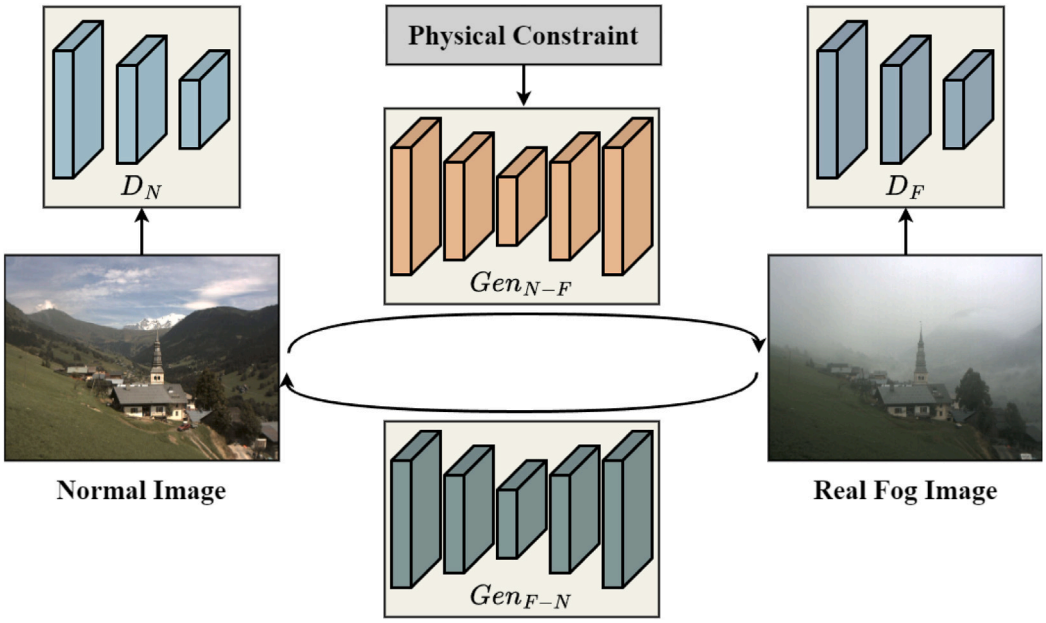

Fig. 5. Illustration of AS-GAN. Gen_{N-F}, Gen_{F-N} denote the generators from Normal image (N) to Real Fog image (F) and F to N, D_N, D_F denote the discriminators for N and F, respectively. Physical Constraint is generated by the atmospheric scattering model, which is imposed on Gen_{N-F}.

Fig. 6. In this study on the impact of fog on images, we analyzed three variants: clean images (row 1), synthetic foggy images generated by the atmospheric scattering model (row 2), and simulated foggy images produced by AS-GAN (row 3). Notably, the simulated foggy images more accurately capture the atmospheric characteristics of real fog compared to the synthetic foggy images.

respectively. y_s serves as the positive example for y, together with the clean data $\hat{y}$, while five other images $(n_1, n_2, n_3, n_4, n_5)$ are randomly chosen from the same batch as negative examples. The equation can be expressed as follows:

$$\mathcal{L}_{CL} = log \frac{sim(y, \hat{y}) + sim(y, y_s)}{sim(y, \hat{y}) + sim(y, y_s) + \sum_{i=1}^{5} sim(y, y_{n_i})}, \tag{24}$$

where sim represents the cosine similarity calculation.

We prompt the synthetic fog domain to progressively close to the real fog domain through this method, thus improving the PED-YOLO's domain adaptability and overcoming performance limitations, as shown in Fig. 4.

3.5. Object detection

In this research, we choose the one-stage detector YOLOv7 as the detector in the detection module, which is the state-of-the-art real-time object detector. Contrasted with earlier versions of YOLO series, the method proposed by YOLOv7 places emphasis on optimizing the training process to improve the detector's accuracy without augmenting inference time consumption. Furthermore, extended and composite scaling techniques are introduced for detector implementation, permitting more adaptable management of parameters and computation. We follow the same network architecture and loss functions present in the original YOLOv7. Finally, we optimize the parameters of the CAP and

Table 2
Statistics of the all datasets.

Dataset	Image number
VOC_Foggy_train	17,709
VOC_Foggy_Simulation	17,709
VOC_Foggy_val	4427
VOC_Foggy_test	4952
RTTS	4432
Foggy_Cityscapes_train	2975
Foggy_Cityscapes_Simulation	2975
Foggy_Cityscapes_val	500
Foggy_Cityscapes_test	1525
Foggy driving	101
MRFID	1000

YOLOv7 by minimizing the following loss:

$$\mathcal{L} = \mathcal{L}_{pip} + \mathcal{L}_{CL} + \mathcal{L}_{Detection}, \tag{25}$$

where $\mathcal{L}_{Detection}$ denotes the object detect loss.

4. Experiments

In this section, we would like to evaluate the performance of PED-YOLO. Extensive experiments on various prevalent datasets under foggy conditions are conducted to explore and address the following research questions (**RQs**):

- **RQ1**: How does the AS-GAN outperform in the task of foggy image generation?
- **RQ2**: How does the PED-YOLO perform in comparison with other real-time methods?
- **RQ3**: Can multi-module fusion actually improve object detection performance?
- **RQ4**: Is PED-YOLO sensitive to various losses?
- **RQ5**: How does the PED-YOLO outperform in efficiency?

4.1. Datasets

VOC (Everingham et al., 2010) is an impressive dataset that contains rich annotation details. It consists of 22,136 images for training and 4952 images for testing. It covers 20 different categories and contains a remarkable 81,640 individual object instances in total. As one of the most widely used benchmark datasets for object detection, this dataset occupies a significant position.

RTTS (Li et al., 2018) is a fairly extensive real-world dataset under foggy circumstances, encompassing 4322 naturally obscured images that span five annotated object categories: person, bicycle, car, bus, and motorcycle. Within the realm of foggy object detection, this dataset is frequently employed to validate model performance.

Foggy Cityscapes (Sakaridis et al., 2018) is a novel dataset derived from the Cityscapes dataset. By incorporating simulated fog effects, it features varying fog concentrations, including dense, medium, and light fog. This approach enables the generation of photorealistic synthetic fog images, which proves beneficial for training and evaluating computer vision algorithms under true-to-life fog conditions. The dataset comprises a total of 5000 images, with 2975 allocated for training, 500 for validation, and 1525 for testing.

Foggy Driving (Sakaridis et al., 2018) is designed for examining and assessing the performance of autonomous driving systems in foggy settings. It encompasses numerous images of road scenes captured under foggy conditions. This dataset comprises more than 500 vehicle instances (including car, truck, bus, train, motorcycle, and bicycle) and approximately 300 human instances (such as person and rider), all annotated from 101 natural fog images.

MRFID (Liu et al., 2020) is a dataset consisting of 200 images featuring natural outdoor scenes, with each scene containing a clean image and four fog images at varying concentrations, specifically light fog, moderate fog, high fog, and extreme fog. In the work, we utilize this dataset to train AS-GAN.

Table 3
Image quality analysis on the fog generation task. Physics model represents the atmospheric scattering model.

Method	PSNR	SSIM	LPIPS
Physics model	18.81	0.80	0.33
Cycle-GAN (Zhu et al., 2017)	19.07	0.81	0.32
AS-GAN	**20.61**	**0.85**	**0.29**

4.2. Implementation details

Data Preparation. We train through these datasets to identify common classes, such as person, car, bus, bicycle, and motorcycle. Each dataset is carefully screened to form the dataset required for our experiments. As VOC dataset covers 20 different categories, we need to select the images in VOC that contain labels of person, car, bus, bicycle, and motorcycle. We then construct the synthetic fog dataset VOC_Foggy, based on the VOC dataset, in line with Eq. (5). The MRFID dataset contains 200 true clean images and the corresponding 800 current fog images distributed over different seasons. We used MRFID to train AS-GAN to learn the atmospheric properties of fog. We set 1000 training iterations to gradually reveal the characteristics of fog through AS-GAN's cyclic learning mechanism (i.e., transformation from normal images to fog images and from fog images back to normal images). We then apply the trained AS-GAN in VOC and Cityscapes dataset to form the VOC_Foggy_Simulation and Foggy_Cityscapes_Simulation datasets. A summary of all data can be found in Table 2.

Training Details. We follow the training protocol as in Wang et al. (2023a) for our proposed method PED-YOLO. During training, we resize the images to 640 × 640. In addition, data augmentation techniques such as image flipping, cropping, and transformations are used to increase the size of the training dataset. PED-YOLO is trained using the Adam optimizer over 80 epochs, with an initial learning rate of 10^{-4} and a batch size of 32. PED-YOLO predicts bounding boxes at three different scales and considers three anchors at each scale. The experiments are conducted in the following environment: PyTorch framework, 1 NVIDIA 3090Ti GPU, and 48 GB of memory.

Competitors and Evaluation Metrics. We compare our approach with baseline IA-YOLOv7, Defog+Detect methods (e.g., MSBDN (Dong et al., 2020) + YOLOv3 (Farhadi and Redmon, 2018), GridDehaze (Liu et al., 2019a) + YOLOv3 (Farhadi and Redmon, 2018), DCPDN (Zhang and Patel, 2018) + YOLOv3 (Farhadi and Redmon, 2018), C^2PNet (Zheng et al., 2023) + YOLOv7 (Wang et al., 2023a), FFA-Net (Qin et al., 2020) + YOLOv7 (Wang et al., 2023a), PSD (Chen et al., 2021a) + YOLOv7 (Wang et al., 2023a)), domain adaptation methods (e.g., DAYOLO (Hnewa and Radha, 2021), MS-DAYOLO (Hnewa and Radha, 2021)), joint learning methods (e.g., DSNet (Huang et al., 2020), IA-YOLO (Liu et al., 2022), IA-YOLOv5, DR-YOLO (Zhong et al., 2024)), and real-time detector CNN-based methods (e.g., YOLOv3 (Farhadi and Redmon, 2018), YOLOv5 (Jocher et al., 2022), YOLOv7 (Wang et al., 2023a)). To evaluate performance, we employ the mean average precision (mAP) at intersection over union (IoU) thresholds of 0.5 and 0.5:0.95, which are the most mainstream evaluation metrics in object detection task.

As part of foggy image generation task, we evaluate the performance of AS-GAN by comparing it with the atmospheric scattering model and Cycle-GAN (Zhu et al., 2017) on the MRFID dataset. Three widely recognized image quality evaluation metrics are employed: PSNR (Wang et al., 2019b), SSIM (Wang et al., 2004), and LPIPS (Zhang et al., 2018). To facilitate this comparison, we use real foggy images as target references, calculating each metric to assess the fidelity of images generated by the atmospheric scattering model, Cycle-GAN, and AS-GAN. In these evaluations, higher PSNR and SSIM values, along with lower LPIPS scores, denote closer alignment between the generated and target images (real foggy images).

Fig. 7. Visualization of real foggy images (row 1), foggy images by the atmospheric scattering model (row 2), foggy images by Cycle-GAN (row 3), and foggy images by AS-GAN (row 4).

*4.3. Fog generation performance (**RQ1**)*

Table 3 demonstrates that AS-GAN outperforms other methods across all three metrics, highlighting its ability to capture the characteristics of real fog. For a more intuitive comparison, we present visualizations of images generated by the three models alongside real foggy images. As shown in Fig. 7, the fog images produced by AS-GAN more closely resemble real foggy images.

*4.4. Overall performance (**RQ2**)*

We comprehensively evaluate the performance of the PED-YOLO model and compare our findings with the performance of current state-of-the-art real-time object detectors such as YOLOv7, C^2PNet+YOLOv7, DSNet, DR-YOLO and IA-YOLO, etc., with respect to four datasets containing two synthetic and two real datasets.

All experiments in Table 4 use VOC_Foggy_train as training data, except our method PED-YOLO, which uses VOC_Foggy_train + VOC_Foggy_Simulation as training data. Additionally, VOC_Foggy_test and RTTS were used as test data for all experiments. In Table 5, all experiments use Foggy_Cityscapes_train as training data, except our method PED-YOLO, which uses Foggy_Cityscapes_train + Foggy_Cityscapes_Simulation. Meanwhile, Foggy_Cityscapes_test and Foggy Driving are used as test data for all experiments. It is noteworthy that while PED-YOLO employs both synthetic and simulated fog data, only synthetic fog data are integrated into the object detector's training pipeline. The simulated fog data are deliberately excluded from the detector's training process for a fair comparison and instead leveraged to strengthen the progressive image processing module's dehazing efficacy, thereby enhancing the framework's generalization performance in real-world fog scenarios.

As shown in Table 4, PED-YOLO achieves state-of-the-art performance with 92.10% $mAP_{0.5}$ and 71.10% $mAP_{0.5:0.95}$ on VOC_Foggy_test and 73.50% $mAP_{0.5}$ and 49.30% $mAP_{0.5:0.95}$ on RTTS. Notably, PED-YOLO outperforms the best baseline model, DR-YOLO, by 1.00%–2.00% across all metrics and surpasses other models by 2.00%–30.00%. The framework demonstrates consistent superiority over joint training methods, domain adaptation approaches, and combined dehazing-detection pipelines, particularly excelling on the RTTS dataset derived from real-world foggy environments.

To further validate PED-YOLO's robustness in foggy traffic scenarios, comparative experiments were performed on the Foggy Cityscapes and Foggy Driving datasets. Table 5 shows that PED-YOLO achieves 60.9% $mAP_{0.5}$ and 36.73% $mAP_{0.5:0.95}$ on Foggy_Cityscapes_test and 58.11% $mAP_{0.5}$ and 34.04% $mAP_{0.5:0.95}$ on Foggy Driving. The model retains a 1.00%–2.00% advantage over DR-YOLO and a 2.00%–24.00%

Table 4
Quantitative evaluations with the state-of-the-art methods on VOC_Foggy_test and RTTS datasets. The best and the second best results (%) are **highlighted** and underlined.

Method	Train data	VOC_Foggy_test		RTTS	
		$mAP_{0.5}$	$mAP_{0.5:0.95}$	$mAP_{0.5}$	$mAP_{0.5:0.95}$
MSBDN (Dong et al., 2020)+YOLOv3 (Farhadi and Redmon, 2018)	VOC_Foggy_train	72.04	57.38	45.90	30.80
GridDehaze (Liu et al., 2019a)+YOLOv3 (Farhadi and Redmon, 2018)	VOC_Foggy_train	72.09	58.23	46.03	31.42
DCPDN (Zhang and Patel, 2018)+YOLOv3 (Farhadi and Redmon, 2018)	VOC_Foggy_train	73.38	58.96	44.43	29.97
DAYOLO (Hnewa and Radha, 2021)	VOC_Foggy_train	66.53	55.11	44.15	29.93
DSNet (Huang et al., 2020)	VOC_Foggy_train	81.71	67.40	49.86	32.03
YOLOv3 (Farhadi and Redmon, 2018)	VOC_Foggy_train	78.87	63.40	49.45	31.80
IA-YOLO (Liu et al., 2022)	VOC_Foggy_train	83.22	67.03	52.36	37.08
YOLOv5 (Jocher et al., 2022)	VOC_Foggy_train	80.40	66.83	50.50	34.62
IA-YOLOv5	VOC_Foggy_train	83.66	67.30	53.60	38.16
YOLOv7 (Wang et al., 2023a)	VOC_Foggy_train	89.40	68.80	63.50	41.10
MS-DAYOLO (Hnewa and Radha, 2021)	VOC_Foggy_train	74.36	60.21	48.86	31.93
FFA-Net (Qin et al., 2020)+YOLOv7 (Wang et al., 2023a)	VOC_Foggy_train	89.65	68.51	65.48	43.42
PSD (Chen et al., 2021a)+YOLOv7 (Wang et al., 2023a)	VOC_Foggy_train	89.81	68.94	66.87	44.83
DR-YOLO (Zhong et al., 2024)	VOC_Foggy_train	90.13	70.05	71.13	47.34
C^2PNet (Zheng et al., 2023)+YOLOv7 (Wang et al., 2023a)	VOC_Foggy_train	89.90	68.20	70.80	45.60
IA-YOLOv7	VOC_Foggy_train	90.30	69.20	70.90	46.70
PED-YOLO	VOC_Foggy_train+VOC_Foggy_Simulation	**92.10**	**71.10**	**73.50**	**49.30**

Table 5
Quantitative Evaluations with the state-of-the-art methods on Foggy_Cityscapes_test and Foggy Driving datasets. The best and the second best results (%) are **highlighted** and underlined.

Method	Train data	Foggy_Cityscapes_test		Foggy driving	
		$mAP_{0.5}$	$mAP_{0.5:0.95}$	$mAP_{0.5}$	$mAP_{0.5:0.95}$
MSBDN (Dong et al., 2020)+YOLOv3 (Farhadi and Redmon, 2018)	Foggy_Cityscapes_train	42.72	25.81	35.84	24.67
GridDehaze (Liu et al., 2019a)+YOLOv3 (Farhadi and Redmon, 2018)	Foggy_Cityscapes_train	42.91	26.05	36.29	24.97
DCPDN (Zhang and Patel, 2018)+YOLOv3 (Farhadi and Redmon, 2018)	Foggy_Cityscapes_train	43.61	26.34	35.16	23.78
DAYOLO (Hnewa and Radha, 2021)	Foggy_Cityscapes_train	40.27	24.37	34.95	23.11
DSNet (Huang et al., 2020)	Foggy_Cityscapes_train	48.91	29.98	40.10	26.50
YOLOv3 (Farhadi and Redmon, 2018)	Foggy_Cityscapes_train	44.66	27.91	38.84	25.52
IA-YOLO (Liu et al., 2022)	Foggy_Cityscapes_train	49.92	30.11	41.21	27.60
YOLOv5 (Jocher et al., 2022)	Foggy_Cityscapes_train	48.60	27.82	39.91	26.03
IA-YOLOv5	Foggy_Cityscapes_train	53.74	32.83	49.97	29.65
YOLOv7 (Wang et al., 2023a)	Foggy_Cityscapes_train	58.10	34.03	55.20	31.91
MS-DAYOLO (Hnewa and Radha, 2021)	Foggy_Cityscapes_train	44.34	26.75	37.79	25.03
FFA-Net (Qin et al., 2020)+YOLOv7 (Wang et al., 2023a)	Foggy_Cityscapes_train	58.13	34.17	55.31	31.94
PSD (Chen et al., 2021a)+YOLOv7 (Wang et al., 2023a)	Foggy_Cityscapes_train	58.18	34.46	55.64	31.99
DR-YOLO (Zhong et al., 2024)	Foggy_Cityscapes_train	59.06	35.00	57.73	33.06
C^2PNet (Zheng et al., 2023)+YOLOv7 (Wang et al., 2023a)	Foggy_Cityscapes_train	58.24	34.61	55.81	32.03
IA-YOLOv7	Foggy_Cityscapes_train	58.40	34.74	57.50	32.93
PED-YOLO	Foggy_Cityscapes_train+Foggy_Cityscapes_Simulation	**60.90**	**36.73**	**58.11**	**34.04**

margin over other models, with pronounced efficacy on the real-world Foggy Driving dataset.

Comprehensive evaluations across multiple benchmarks confirm PED-YOLO's exceptional performance, particularly its dominance in authentic foggy conditions compared to existing state-of-the-art models. While not yet integrated into autonomous driving systems, our rigorous experimentation addresses critical safety requirements for object detection in foggy environments. The results highlight our proposal's improved detection accuracy under real-world foggy conditions, demonstrating its potential to enhance perception reliability for autonomous vehicles and thereby advance operational safety.

*4.5. Ablation study on components (**RQ3**)*

We investigate the effectiveness of various components in our proposed PED-YOLO, including the Progressive Image Processing module (PIP), the Channel Adaptive Predictor (CAP), and the simulated data Contrastive Learning (SCL). Our **base** network is IA-YOLOv7, and we subsequently develop six variants as follows: (1) base+PIP: Incorporating PIP into the image adaptive module. (2) base+CAP: Replacing the original image processing module with the whole module CAP. (3) base+SCL : Adding simulated data contrastive learning to the base, with a 2:5 ratio between positive and negative samples. (4) base+PIP+SCL: Integrating simulated data contrastive learning with base+PIP while maintaining a 2:5 ratio between positive and negative samples. (5) base+CAP+SCL: Integrating simulated data contrastive learning with base+CAP while maintaining a 2:5 ratio between positive and negative samples. (6) PED-YOLO: The comprehensive model of our PED-YOLO. We showcase the results in Table 6, using VOC_Foggy_train for training and VOC_Foggy_test and RTTS for testing.

The PIP module addresses the challenge of filtering large-sized foggy images at once and follows the "coarse to fine" principle by applying filters at different scales. The CAP module focuses on important information in each channel of the image for more comprehensive adaptive enhancement. The SCL module uses a contrast learning paradigm to push the synthetic fog domain of VOC_Foggy to be closer to the real fog domain. As shown in Table 6, both the PIP and CAP modules can improve object detection by about 1 to 2%. However, applying the SCL method to base alone results in a drop in performance, while applying the SCL method with base+PIP and base+CAP results in performance improvements. This suggests that the base does not fully exploit the potential of contrast learning. For all other variants, contrast

Table 6
Ablation study on PED-YOLO with different modules on VOC_Foggy_test and RTTS datasets. Results (%) from the baseline model are underlined in the experimental data, while data from our proposed method are highlighted in bold.

Model	Train data	VOC_Foggy_test		RTTS	
		$mAP_{0.5}$	$mAP_{0.5:0.95}$	$mAP_{0.5}$	$mAP_{0.5:0.95}$
base	VOC_Foggy_train	90.30	69.20	70.90	46.70
base+PIP	VOC_Foggy_train	91.50	69.50	71.90	48.30
base+CAP	VOC_Foggy_train	91.30	69.60	71.40	47.50
base+SCL	VOC_Foggy_train+VOC_Foggy_Simulation	90.90	69.30	69.90	45.80
base+PIP+SCL	VOC_Foggy_train+VOC_Foggy_Simulation	91.50	70.20	72.70	48.50
base+CAP+SCL	VOC_Foggy_train+VOC_Foggy_Simulation	91.30	69.40	73.20	48.90
PED-YOLO	VOC_Foggy_train+VOC_Foggy_Simulation	**92.10**	**71.10**	**73.50**	**49.30**

Table 7
Ablation study on losses in progressive image processing module. (%)

Losses	VOC_Foggy_test		RTTS	
	$mAP_{0.5}$	$mAP_{0.5:0.95}$	$mAP_{0.5}$	$mAP_{0.5:0.95}$
$\mathcal{L}_{MSE}$	91.50	69.40	72.10	48.50
+ $\mathcal{L}_{tv}$	91.50	69.50	72.20	48.70
+ $\mathcal{L}_{color}$	91.50	69.50	72.10	48.80
+ $\mathcal{L}_{dcp}$	91.70	69.40	72.20	48.90
+ $\mathcal{L}_{per}$	91.90	70.60	72.10	49.10
+ $\mathcal{L}_{rec}$	92.10	71.10	73.50	49.30

Table 8
Efficiency analysis on the compared methods.

Method	Params	Speed (ms)
YOLOv7	74.8M	3.1
IA-YOLOv7	75.1M	4.3
C^2PNet+YOLOv7	113.9M	45.3
DETR-Res50 (Carion et al., 2020)	159.0M	83.0
DETR-Res101 (Carion et al., 2020)	232.0M	97.2
PED-YOLO	75.2M	7.1

learning contributes to improved performance. PED-YOLO combines all modules to achieve optimal results, and the effectiveness of our proposed module is suggested by the ablation experiment results.

4.6. Ablation study on losses (*RQ4*)

In the progressive image processing module, we design a multi-scale loss framework incorporating six distinct loss functions (mean squared error (MSE) loss $\mathcal{L}_{MSE}$, reconstruction loss $\mathcal{L}_{rec}$, perceptual loss (Johnson et al., 2016) $\mathcal{L}_{per}$, dark channel prior (DCP) loss (He et al., 2010) $\mathcal{L}_{dcp}$, color loss $\mathcal{L}_{color}$, and total variation loss $\mathcal{L}_{tv}$) to optimize the dehazing capability, thereby enhancing subsequent object detection performance. Table 7 presents ablation studies demonstrating the incremental contributions of different loss functions through progressive addition starting with the MSE loss $\mathcal{L}_{MSE}$, evaluated on both VOC_Foggy_test and RTTS datasets.

Adding the total variation loss $\mathcal{L}_{tv}$ yields a 0.1% improvement in $mAP_{0.5:0.95}$ on VOC_Foggy_test and marginal gains on RTTS. By suppressing noise-induced local pixel variations, the TV loss enhances spatial smoothness in denoised images, thereby moderately improving detection robustness. Adding the color loss $\mathcal{L}_{color}$ achieves a 0.1% enhancement in $mAP_{0.5:0.95}$ on RTTS, indicating its effectiveness in mitigating color distortion from foggy conditions through chromatic consistency constraints. Adding the dark channel prior loss $\mathcal{L}_{dcp}$ produces more substantial improvements, 0.2% gain in $mAP_{0.5}$ on VOC_Foggy_test and 0.1% increase in $mAP_{0.5:0.95}$ on RTTS. This loss mechanism strengthens global contrast restoration through enhanced haze removal, significantly improving object visibility in foggy scenarios.

Adding the perceptual loss $\mathcal{L}_{per}$ generates the most pronounced improvements, 1.2% enhancement in $mAP_{0.5:0.95}$ on VOC_Foggy_test and 0.2% in $mAP_{0.5:0.95}$ on RTTS. By enforcing high-level semantic feature constraints, this loss preserves critical structural information relevant to object detection, enhancing model adaptability to complex weather interference. Adding the reconstruction loss $\mathcal{L}_{rec}$ achieves 0.2%, 0.5% improvements in $mAP_{0.5}$ and $mAP_{0.5:0.95}$ on VOC_Foggy_test, and a notable 1.4% gain in $mAP_{0.5}$ on RTTS through pixel-level fidelity optimization.

In summary, experiments demonstrate that the synergistic operation of multiple losses achieves balanced image detail recovery and semantic consistency, ultimately enhancing object detection robustness under foggy conditions.

4.7. Efficiency analysis (*RQ5*)

We provide a detailed analysis of our proposed method, PED-YOLO, focusing on two critical metrics: the parameter set and inference time. To prove the effectiveness of PED-YOLO, we compare it with current state-of-the-art object detection algorithms DETR (Carion et al., 2020), as illustrated in Table 8.

As shown in Table 8, the DETR method is time-consuming and the model parameters are relatively large, so it is less suitable for autonomous driving scenarios with high real-time requirements. In contrast, PED-YOLO demonstrates its efficacy by reasoning a 640 × 640x3 image in merely 7.1 ms on a single 3090Ti GPU. Although its efficiency is not the highest, lagging behind YOLOv7 and IA-YOLOv7 by 4.0 ms and 2.8 ms respectively, the marginal increase in its parameter count, by only 0.4M and 0.1M respectively, is notable. This discrepancy is primarily due to PED-YOLO containing more intensive image computation steps, thereby extending its inference time, while keeping the parameter increase minimal. Despite not being the most efficient, PED-YOLO satisfies real-time processing requirements and surpasses other methods by 2%–8% in the $mAP_{0.5}$ and $mAP_{0.5:0.95}$.

Moreover, for comparative analysis and result visualization, we compare YOLOv7 and its enhanced version, IA-YOLOv7, with PED-YOLO. As illustrated in Fig. 8, our model outperforms others, achieving superior disarming outcomes, greater accuracy, more detections, and enhanced defogging capabilities. Despite these improvements, our model faces limitations, especially in defogging efficacy. The depiction of sky scenes requires additional refinement.

5. Conclusion

We propose PED-YOLO, a novel framework aimed at enhancing object detection in foggy conditions for autonomous driving systems. Each input image is adaptively processed to optimize detection accuracy. Our method introduces a fully differentiable Progressive Image Processing module (PIP), whose hyperparameters are dynamically predicted by a Channel Adaptive Predictor (CAP). This module eliminates weather-specific artifacts while enabling the YOLOv7 detector to identify latent object features. PED-YOLO implements a coarse-to-fine filtering strategy to address challenges in processing high-resolution foggy images. Furthermore, we propose AS-GAN, a style transfer model that generates simulated fog data closely approximating real-world conditions. To enhance generalization, PED-YOLO incorporates contrastive learning by leveraging synthetic and simulated fog datasets. Experimental results show that PED-YOLO surpasses state-of-the-art methods in foggy

Fig. 8. Detection results of YOLOv7 (row 1), baseline IA-YOLOv7 (row 2) and our PED-YOLO (row 3). We label the significant difference (as shown in red boxes), and enlarge it on the right side of each image. PED-YOLO can reduce the fog and sharpen image edge, which has better detection performance with fewer missed and wrong detections.

environments. We assert that this work contributes to the development of robust perception systems for autonomous vehicles.

As demonstrated in Fig. 8, PED-YOLO achieves significant performance improvements. However, limitations remain in the dehazing process, particularly in rendering sky regions, which appears somewhat unnatural. Addressing this challenge will be a primary focus of future investigations. Concurrently, multimodal research (Wang et al., 2022, 2023b) is rapidly advancing, as evidenced by autonomous vehicles integrating data from radar, LiDAR, and meteorological sources. Our goal is to unify these multimodal inputs to enhance detection accuracy in adverse weather. Projected advancements in multimodal perception could significantly improve system robustness, constituting a primary goal of subsequent studies. We also aim to broaden PED-YOLO's applicability by simulating diverse adverse weather scenarios, including rain and hail.

CRediT authorship contribution statement

Zhiying Li: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Junhao Wu:** Software, Methodology, Investigation, Formal analysis. **Shuyuan Lin:** Writing – review & editing, Visualization. **Zheng Wang:** Writing – review & editing, Visualization. **Xiaobo Jin:** Writing – review & editing. **Guanggang Geng:** Writing – review & editing, Supervision. **Feiran Huang:** Writing – review & editing, Supervision. **Jian Weng:** Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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